Covariance Matrix Accelerator

Single-Pass RTL Design

EE3510 – RTL Design and Verification IIT Hyderabad

1 Abstract

This report describes a parameterized SystemVerilog implementation of a single-pass covariance-matrix accelerator. For the tested configuration, the design accepts four samples, each containing three signed 16-bit dimensions. It accumulates the cross-products required for the matrix $X^T X^T$ and the sum of each dimension. The accumulated values are serialized as twelve 32-bit words: nine cross-product sums followed by three dimension sums. A shared integer-to-binary32-style conversion function is used on the output path.

2 Mathematical Background

For a dataset with N samples and D dimensions, let $x_{n,i}$ be dimension i of sample n. The population covariance between dimensions i and j is

$$\mathrm{Cov}(i,j) = \frac{1}{N}\sum_{n=0}^{N-1} x_{n,i}x_{n,j} - \frac{1}{N^2}\left(\sum_{n=0}^{N-1} x_{n,i}\right)\left(\sum_{n=0}^{N-1} x_{n,j}\right).$$

Equivalently, define

$$S_i = \sum_{n=0}^{N-1} x_{n,i}, \qquad P_{i,j} = \sum_{n=0}^{N-1} x_{n,i}x_{n,j}.$$

Then

$$\mathrm{Cov}(i,j) = \frac{P_{i,j} - (S_i S_j/N)}{N}.$$

This identity allows the design to accumulate the dimension sums S_i and cross-product sums $P_{i,j}$ during one pass over the samples. The testbench performs the final covariance calculation from the serialized accumulations. The RTL outputs the accumulations rather than directly outputting the final covariance matrix.

2.1 Output encoding

Each output word is 32 bits and is intended to represent a single-precision-style floating-point value. The conversion function determines the sign and leading-one position, forms the biased exponent using 127, and extracts a 23-bit mantissa field.

3 Design Specification

The RTL module's parameters and interface are shown below.

Parameter	Tested value
`DATA_WIDTH`	16
`DIMENSIONS`	3
`SAMPLES`	4
Clock period in testbench	10 ns

The principal inputs are `clk`, `rst`, `start`, signed `data_in`, and `data_valid`. The outputs are `done`, `result_out[31:0]`, `result_index[4:0]`, and `result_valid`.

4 Control FSM

The RTL defines four states: `IDLE`, `PROCESS_DATA`, `PROCESS_RESULTS`, and `DONE_STATE`.

- **IDLE:** waits for `start` and clears the result counter.
- **PROCESS_DATA:** accepts input values when `data_valid` is asserted. A dimension counter identifies the position within each sample, and a sample counter tracks completed samples.
- **PROCESS_RESULTS:** asserts `result_valid` and serializes the nine entries of P, followed by the three entries of S.
- **DONE_STATE:** asserts `done`; the next-state logic returns to idle when `start` is low.

5 Datapath and Output Sharing

During data intake, the design stores the current sample's dimensions in `current_sample_reg`. When the last dimension of a sample arrives, nested loops update all cross-product accumulators:

$$P_{i,j} \leftarrow P_{i,j} + x_i x_j, \qquad S_i \leftarrow S_i + x_i.$$

The last incoming dimension is taken directly from `data_in`; earlier dimensions are taken from the sample registers. This allows the final dimension to participate in the same update cycle in which it is received.

The output logic selects one accumulated integer based on `result_count`. A single conversion function, `f_signed_to_float_SHARED`, converts the selected value to the 32-bit output format. This is the output-sharing optimization reflected in the module name and comments. The serialized interface uses one 32-bit output bus rather than exposing all accumulators simultaneously.

Listing 1: Representative output-selection logic from the RTL

```
always_comb begin
  if (result_count < DIMENSIONS*DIMENSIONS)
    current_int_to_convert =
      XX_T_sum[result_count / DIMENSIONS]
              [result_count % DIMENSIONS];
  else
    current_int_to_convert =
```

```
      dim_sum[result_count - DIMENSIONS*DIMENSIONS];
end

assign result_out =
  f_signed_to_float_SHARED(current_int_to_convert);
assign result_index = result_count;
```

6 Functional Verification

The supplied testbench uses $D = 3$, $N = 4$, and the input matrix

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & 7 & 11 \\ 4 & 8 & 12 \end{bmatrix}.$$

It independently calculates expected dimension sums and cross-products, computes the reference covariance matrix, captures the DUT's serialized values, and compares the resulting covariance entries using $\epsilon = 10^{-4}$.

The expected accumulated values are

$$P = \begin{bmatrix} 27 & 55 & 83 \\ 55 & 115 & 175 \\ 83 & 175 & 267 \end{bmatrix}, \qquad S = \begin{bmatrix} 9 & 17 & 25 \end{bmatrix}.$$

The expected population covariance matrix is

$$C = \begin{bmatrix} 1.6875 & 4.1875 & 6.6875 \\ 4.1875 & 10.6875 & 17.1875 \\ 6.6875 & 17.1875 & 27.6875 \end{bmatrix}.$$

Below are the Tcl Console output and the Simulation result:

```
----------------------------------------
Starting Covariance Matrix Testbench
DATA_WIDTH=16, DIMENSIONS=3, SAMPLES=4
----------------------------------------
Sending input samples...

SUCCESS: Test Passed! All results match.

--- Covariance Matrix ---
[    1.68750    4.18750    6.68750 ]
[    4.18750   10.68750   17.18750 ]
[    6.68750   17.18750   27.68750 ]

TEST COMPLETED
```

Figure 1: Vivado TCL Console output

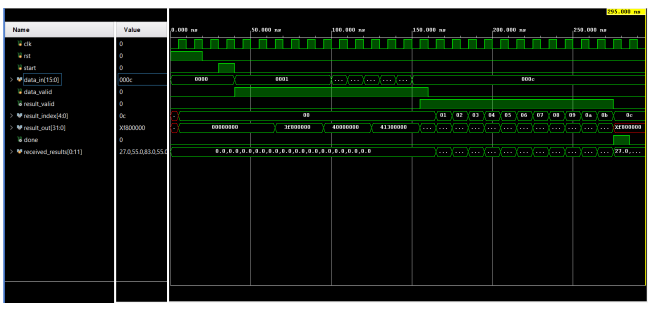


Figure 2: Vivado Simulation waveform.

7 Implementation Results

The following metrics are transcribed from the implemented-design.

7.1 Resource utilization

Resource	Used	Available
CLB LUTs	1,496	230,400
CLB registers	1,207	460,800
CARRY8	14	28,800
F7 muxes	2	115,200
CLBs	379	28,800
LUTs as logic	1,398	230,400
LUTs as memory	98	101,760
DSPs	9	1,728
Global clock buffers	1	544
PS8	1	1

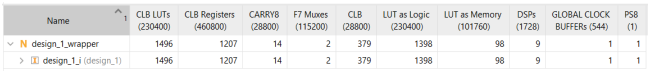


Figure 3: Vivado implemented-design utilization hierarchy.

7.2 Timing analysis

The implemented timing summary reports:

- Worst Negative Slack (WNS): +3.477 ns.
- Total Negative Slack (TNS): 0.000 ns.
- Setup failing endpoints: 0.
- Worst Hold Slack (WHS): +0.013 ns.
- Total Hold Slack (THS): 0.000 ns.
- Hold failing endpoints: 0.
- Worst Pulse Width Slack: +3.500 ns.
- Pulse-width failing endpoints: 0.

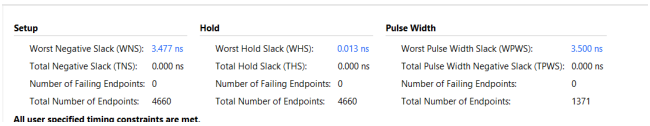


Figure 4: Implemented-design timing summary.

7.3 Power analysis

The implemented power summary reports total on-chip power of 3.353 W, consisting of 2.661 W dynamic power (79%) and 0.691 W static power (21%).

The PS accounts for 2.637 W of the reported dynamic power. The report also lists 0.007 W for clocks, 0.009 W for signals, 0.007 W for logic, and 0.001 W for DSPs. Junction temperature is reported as 28.3°C, and the confidence level is medium. Because the processing system dominates the estimate, the total 3.353 W is reported for

the implemented design and is not attributed solely to the covariance accelerator.

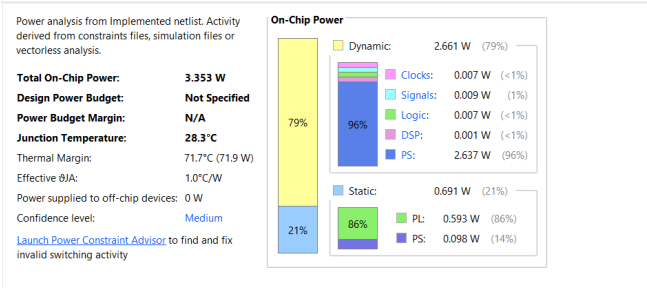


Figure 5: Vivado implemented-design power summary.

8 Conclusion

The supplied RTL and testbench demonstrate a single-pass covariance-accumulation architecture for the tested configuration of three dimensions and four samples. The design accumulates the cross-product matrix and per-dimension sums, then serializes these twelve values through a shared conversion function. The provided simulation output reports that the covariance reconstructed by the testbench matches the reference result within the specified tolerance.

The supplied implementation reports show 1,496 CLB LUTs, 1,207 CLB registers, and 9 DSPs, with no reported setup, hold, or pulse-width timing failures. The total on-chip power estimate is 3.353 W, dominated by the Processing System. These hardware figures describe the reported ZCU104 implemented design.